AI-Based Archaeological Pottery Fragment Matching

A computer-vision approach to matching fragments and exploring digital reconstruction



PRT LLT

Review 1

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PROBLEM

Thousands of Fragments, One Lost Object

Pottery fragments are weathered, incomplete, and often look alike

Manual sorting and matching take significant expert time

The process doesn’t scale as collections grow

Our goal: rank plausible matches to support experts, not replace them



Objectives and Pattern Recognition Task

What we take in:
an image of a pottery fragment

What we return:
ranked candidate fragments
likely to fit

What we look for:
shape, texture, curvature,
and fracture-edge geometry

How we measure success:
high-quality retrieval that supports experts —
not autonomous reconstruction





DATASET

Dataset and Data Understanding



We will evaluate the daxinzhuang pottery-fragment dataset available via zenodo record

Inspect image count, views, resolution, labels, and train/test split before finalizing

Dataset documentation may differ — verify fields and annotations carefully

Exact downloadable contents and annotations must be verified

EXAMPLE DATA SCHEMA (VERIFY BEFORE USE)

Fragment ID	Image / View	Match Label (if supplied)	Provenance (if available)	Context / Site (if available)	Notes (if available)
Unique identifier for each fragment	View index or filename	Match / non-match or fragment group	Collection, site, or excavation info	Stratography, layer, or context details	Any additional observations

METADATA

Metadata
tells us what pixels cannot

Metadata describes the sample's **identity and context**

Image pixels provide **visual evidence**

Reliable metadata enables confident **matching** and **prevents data leakage**

Split by original vessel or fragment group to **avoid near-duplicate leakage**

METADATA

Fragment identifier

Excavation / provenance context

View type

Expert match annotation

+

PIXELS

Texture, colour, shape, pattern and other visual features



THE PROCESS

Preprocessing Pipeline

Clean, standardize, and capture the fragments’ true geometry.

01

Validate and deduplicate



Remove corrupt files and duplicates, ensure reliable inputs.

02

Standardize images



Consistent size, orientation, and color handling.

03

Isolate fragment from background



Separate the fragment to focus on shape and surface.

04

Normalize illumination



Reduce lighting variations while preserving detail.

05

Extract edges and contour



Capture fracture edges and contours for matching.

PROPOSED METHODOLOGY

AI-BASED ARCHAEOLOGICAL POTTERY FRAGMENT MATCHING

FRAGMENT IMAGE

VISUAL FEATURES

CANDIDATE RETRIEVAL

GEOMETRIC COMPATIBILITY SCORING

RANKED MATCHES

VIRTUAL ASSEMBLY (OPTIONAL)



BASELINE

Content-based retrieval using appearance similarity

Ranking by visual closeness only

No geometric reasoning or assembly

PROPOSED APPROACH

Deep visual features for robust retrieval

Geometric compatibility scoring for precise alignment

Ranked matches to support accurate pairing

FUTURE EXTENSION

Virtual reassembly of matched fragments

Context-aware refinement with iterative feedback

Scalable integration into digital workflows

PLANNED METHODOLOGY; NO EXPERIMENTAL RESULTS YET

THE PROBLEM

Research Gap and Hypothesis

Does adding fracture-edge geometry
improve top-k matching vs. image-only retrieval?

BASELINE

Image-only retrieval

appearance-based similarity

PROPOSED

Combined scoring

appearance + fracture-edge geometry

EVALUATE WITH	Precision	Recall	Mean reciprocal rank where labels permit

We will **check related work** before making originality claims.



Scope, Impact, Limitations and References

PRACTICAL VALUE

Prioritizes the most plausible matches to save expert time

Supports consistent documentation and digital archives

KEY LIMITATIONS

Erosion and surface wear

Missing fragments

Lighting and imaging variability

Ambiguous edges and incomplete labels

NEXT STEPS

Verify dataset access and labels

Review relevant literature

Implement a baseline model

Evaluate on held-out fragments

IMPACT

Improves triage and decision quality

Enables scalable analysis as collections grow

Strengthens preservation and research outcomes



REFERENCES

Dataset record: [Dataset Name], [Repository or Institution], [Accessed Date], [DOI or URL]

Pottery reassembly approach and evaluation: [Author(s}],. [Title], [Journal/Conference], [Year], [Volume(issue)]:[Pages]. [DOI]

Fragment matching with deep learning: [Author(s}],. [Title], [Journal/Conference], [Year], [Volume(issue)]:[Pages]. [DOI]